

ALEKSANDROVSKIY ZAVOD, Russian Federation

WMO: 309710

Lat: 50.917N

Lon: 117.933E

Elev: 807

StdP: 92.00

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.6	(c) -36.1	(d) -41.5	(e) 0.1	(f) -38.4	(g) -38.8	(h) 0.1	(i) -35.9	(j) 7.9	(k) -15.9	(l) 6.9	(m) -18.9	(n) 1.0	(o) 200	(p) 0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	28.5	17.4	26.6	17.0	24.8	16.6	19.5	25.3	18.6	24.0	17.7	22.8	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.9	22.2	16.7	13.1	20.9	15.7	12.3	19.9	59.5	25.3	56.4	24.1	53.3	22.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.6	(b) 7.4	(c) 6.4	(e) -41.2	(f) 31.9	(g) 2.3	(h) 2.4	(i) -42.9	(j) 33.6	(k) -44.2	(l) 35.0	(m) -45.5	(n) 36.3	(o) -47.2	(p) 38.1		
(a) 8.6	(b) 7.4	(c) 6.4	(e) -41.3	(f) 21.4	(g) 2.3	(h) 0.8	(i) -42.9	(j) 22.0	(k) -44.3	(l) 22.4	(m) -45.6	(n) 22.8	(o) -47.2	(p) 23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-25.0	-21.6	-12.1	-0.5	8.0	14.6	17.1	14.8	7.9	-2.0	-15.0	-23.5
	DBStd	15.91	4.54	5.24	6.79	5.08	4.26	3.66	2.79	3.54	4.33	5.39	6.41	4.87
	HDD10.0	5325	1085	886	685	317	90	6	1	6	89	373	750	1037
	HDD18.3	7827	1343	1119	943	566	323	121	55	118	312	631	1000	1296
	CDD10.0	575	0	0	0	1	27	143	222	154	27	1	0	0
	CDD18.3	36	0	0	0	0	1	8	18	7	1	0	0	0
	CDH23.3	744	0	0	0	1	33	239	300	149	22	0	0	0
	CDH26.7	173	0	0	0	0	5	68	68	29	3	0	0	0
Wind	WSAvg	2.3	1.5	1.8	2.5	3.2	3.4	2.5	2.2	2.2	2.4	2.5	2.0	1.5
Precipitation	PrecAvg	383	4	4	7	12	33	67	96	91	40	13	8	5
	PrecMax	603	12	15	18	37	91	210	192	208	138	42	25	16
	PrecMin	247	0	0	2	1	1	19	12	26	1	0	0	1
	PrecStd	86	3	3	5	10	24	41	41	44	33	10	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.0	-4.4	10.2	20.5	27.2	32.2	31.3	29.7	26.4	17.2	4.9	-5.8
		MCWB	-9.6	-6.6	2.6	9.3	14.1	17.3	19.0	18.6	14.4	8.4	0.5	-7.4
	2%	DB	-12.8	-7.3	5.7	16.1	23.5	28.2	28.4	26.9	22.7	12.7	0.8	-11.1
		MCWB	-13.6	-8.9	0.0	6.7	12.3	16.7	18.3	17.4	12.9	5.4	-2.1	-12.1
	5%	DB	-15.0	-9.8	2.7	12.6	20.8	25.9	26.5	24.8	19.7	9.9	-2.3	-13.5
		MCWB	-15.7	-11.0	-1.6	4.6	10.7	15.8	18.1	17.1	11.4	4.0	-4.2	-14.2
	10%	DB	-17.2	-11.9	-0.5	9.5	18.1	23.7	24.7	22.7	17.1	6.9	-4.9	-15.7
		MCWB	-17.8	-12.9	-3.5	2.9	9.2	15.0	17.6	16.2	10.4	2.4	-6.3	-16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.6	-6.5	3.0	9.3	15.0	19.6	21.1	20.3	15.5	8.3	0.8	-7.3
		MCDB	-8.9	-4.5	9.2	19.9	25.2	29.1	27.2	26.7	22.5	16.4	4.3	-5.8
	2%	WB	-13.5	-8.8	0.5	6.8	13.1	18.2	19.9	18.8	14.1	6.1	-1.9	-12.0
		MCDB	-12.9	-7.3	5.3	15.6	21.7	25.2	25.4	24.2	19.6	11.1	0.5	-11.2
	5%	WB	-15.7	-10.9	-1.4	4.9	11.4	17.1	19.1	17.8	12.8	4.4	-4.2	-14.2
		MCDB	-15.0	-9.8	2.6	12.3	19.4	23.2	24.2	23.0	18.0	9.3	-2.4	-13.6
	10%	WB	-17.7	-12.9	-3.5	3.1	9.7	16.1	18.2	16.8	11.2	2.8	-6.3	-16.2
		MCDB	-17.2	-11.9	-0.6	9.0	17.1	21.7	23.2	21.3	16.4	6.6	-4.9	-15.7
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.5	16.0	14.3	15.2	15.0	13.4	13.9	14.9	13.8	13.2	12.8
		MCDBR	14.7	16.1	17.2	20.3	20.5	19.9	17.6	18.0	19.6	17.4	14.7	14.0
		MCWBR	13.9	14.7	12.3	12.2	10.9	9.6	9.0	9.5	10.6	11.1	11.7	13.0
	5% WB	MCDBR	14.6	15.9	16.7	19.0	18.2	15.9	13.4	14.1	16.0	16.2	14.3	13.9
		MCWBR	13.9	14.5	12.2	11.7	10.3	8.4	7.6	8.1	9.1	10.6	11.6	12.9
Clear-Sky Solar Irradiance	taub	0.247	0.273	0.320	0.416	0.484	0.465	0.416	0.386	0.344	0.313	0.276	0.254	
	taud	2.219	2.176	2.114	2.004	1.885	2.053	2.265	2.323	2.378	2.362	2.265	2.175	
	Ebn at Noon	786	856	870	816	774	797	834	843	844	803	736	701	
	Edh at Noon	69	96	124	158	187	159	127	114	96	80	66	62	
All-Sky Solar Radiation	RadAvg	1.56	2.72	4.26	5.25	5.89	6.15	5.66	5.05	3.97	2.70	1.58	1.17	
	RadStd	0.06	0.11	0.20	0.29	0.43	0.56	0.38	0.37	0.29	0.17	0.07	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	-1.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page